

Smart Glass Solutions for Commercial, Residential, Wellness and Luxury Applications



Smart glass is an advanced glazing technology that allows glass surfaces to switch between transparent, translucent, and opaque states. It provides privacy on demand, energy efficiency, solar control, modern aesthetics, and smart-building integration.

Main Technologies

PDLC (Privacy Glass),SPD (Dynamic Tint Glass), Electrochromic Glass, Thermochromic Glass, and Photochromic Glass.

Technology	Brief Description
PDLC (Privacy Glass)	Uses liquid crystal technology to switch instantly between transparent and opaque states for privacy on demand.
SPD (Dynamic Tint Glass)	Uses suspended particles to adjust the glass tint level, reducing glare and solar heat while maintaining visibility.
Electrochromic Glass	Changes tint gradually through an electrical charge, providing energy savings and solar control for large glazing areas.
Thermochromic Glass	Automatically darkens when exposed to higher temperatures, reducing heat gain without electricity.
Photochromic Glass	Responds to sunlight and UV exposure by darkening automatically, similar to transition eyeglasses.

Commercial Applications

Conference rooms, executive offices, hospitals, hotels, retail stores, airports, educational facilities, and luxury developments.

Residential Applications

Bathrooms, home offices, living areas, skylights, luxury villas, penthouses, and smart homes.

Luxury Wellness & Fitness Applications

Private gyms, wellness centers, indoor pools, spa suites, hammams, massage rooms, hydrotherapy zones, and resort facilities.

Advantages

Instant privacy, premium aesthetics, smart-home integration, reduced need for blinds, improved hygiene, enhanced property value, and energy savings.

Disadvantages

Higher initial investment, electrical requirements for some technologies, and specialized installation requirements.

Recommended Technologies

PDLC for privacy applications, Electrochromic for façades and skylights, SPD for luxury transportation and premium solar control.





Suggested Image Placement

- Cover Page – Luxury smart glass office or villa
- PDLC transparent vs opaque demonstration
- Executive boardroom installation
- Hospital patient room application
- Luxury hotel suite and bathroom partition
- Home gym with smart glass walls
- Indoor swimming pool and spa area
- Luxury wellness suite / hammam
- Electrochromic building façade
- Before-and-after transformation examples